

Data Quality Monitoring System

Project Report

1. Introduction

Data quality is a critical factor in data analysis and business decision-making. Inaccurate, incomplete, or duplicate data can lead to misleading insights and poor business outcomes. The Data Quality Monitoring System was developed to identify and monitor common data quality issues within a dataset and provide a centralized dashboard for tracking data health.

2. Objective

The main objectives of this project are:

- Detect missing values in critical fields.
 - Identify duplicate records.
 - Validate data based on predefined business rules.
 - Monitor overall data quality using KPIs.
 - Create an interactive dashboard for data quality analysis.
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3. Problem Statement

Organizations often deal with large volumes of data collected from multiple sources. This data may contain errors such as missing values, duplicate records, invalid entries, and inconsistent formats. These issues affect reporting accuracy and decision-making.

The goal of this project is to build a monitoring system that automatically detects data quality issues and provides actionable insights through visual dashboards.

4. Dataset Description

The dataset contains customer-related information with the following fields:

Column Name	Description
Customer ID	Unique identifier for customers
Name	Customer name
Age	Customer age
Gender	Customer gender
Email	Customer email address

Column Name	Description
City	Customer location
Purchase Amount	Transaction amount

5. Data Quality Checks Performed

Missing Value Check

Identifies blank or null values in important columns.

Duplicate Record Check

Detects duplicate Customer IDs and duplicate email addresses.

Age Validation

Flags records where:

- Age is missing
- Age is less than 0
- Age is greater than 120

Email Validation

Verifies whether email addresses follow a valid format.

Purchase Amount Validation

Detects negative or missing purchase values.

Error Flag Generation

Creates a final status column indicating whether a record contains any data quality issue.

6. Dashboard KPIs

The dashboard tracks the following metrics:

- Total Records
- Clean Records
- Error Records
- Missing Values Count
- Duplicate Records Count
- Invalid Age Records
- Invalid Email Records
- Data Quality Score

7. Tools and Technologies Used

- Microsoft Excel
 - Excel Formulas
 - Conditional Formatting
 - Pivot Tables
 - Charts and Visualizations
 - Dashboard Design Techniques
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8. Methodology

1. Collect raw customer data.
 2. Apply validation rules using Excel formulas.
 3. Identify missing, duplicate, and invalid records.
 4. Generate quality indicators and error flags.
 5. Create summary tables using Pivot Tables.
 6. Design an interactive dashboard for monitoring KPIs.
 7. Analyze data quality trends and generate insights.
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9. Results

The system successfully identified data quality issues including:

- Missing customer information
- Duplicate customer records
- Invalid age values
- Incorrect email formats
- Invalid purchase amounts

The dashboard provides a quick overview of data quality status and helps users take corrective actions efficiently.

10. Business Impact

- Improves data accuracy and reliability.
 - Reduces manual data validation efforts.
 - Enhances reporting quality.
 - Supports better business decisions.
 - Establishes a proactive data quality monitoring process.
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11. Future Enhancements

- Power BI Integration

- SQL-Based Data Validation Framework
 - Python Automation for Quality Checks
 - Automated Email Alerts
 - Real-Time Data Monitoring
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12. Conclusion

The Data Quality Monitoring System provides an effective solution for detecting and monitoring data quality issues. By automating validation checks and visualizing quality metrics through dashboards, organizations can improve data reliability, reduce errors, and make more informed decisions.

Author

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Skills Demonstrated:

Excel | Data Cleaning | Data Validation | Dashboard Development | KPI Reporting | Data Analysis